

Hardening and Securing an Operating System

Minor Project Report

Submitted By

AADITYA MISTRY

Submitted To

SmartED Innovations

**Department of Computer Engineering
Sardar Vallabhbhai Patel Institute of Technology
(SVIT)
Vasad, Gujarat**

Submitted in July 2026

CERTIFICATE

This is to certify that Mr. Aaditya Mistry has successfully completed the Minor Project entitled "Hardening and Securing an Operating System" in partial fulfillment of the requirements of the Cybersecurity course conducted by SmartED Innovations.

The project demonstrates the practical implementation of operating system hardening techniques using Windows 11. Various security mechanisms, including Windows Update management, Microsoft Defender configuration, Windows Defender Firewall verification, user account security, NTFS permission analysis, ransomware protection, network security assessment, event logging, and system recovery mechanisms, were implemented and documented.

To the best of our knowledge, this report represents the original work carried out by the student and has not been submitted previously for any other academic purpose.

Submitted To
SmartED Innovations

Department of Computer Engineering
Sardar Vallabhbhai Patel Institute of Technology (SVIT)

Date: July 2026

ACKNOWLEDGEMENT

I would like to express my sincere gratitude to SmartED Innovations for providing me with the opportunity to undertake this Minor Project titled "Hardening and Securing an Operating System." This project has provided me with valuable practical exposure to operating system security, Windows system administration, and cybersecurity best practices.

I am thankful to the mentors and faculty members of SmartED Innovations for their continuous guidance, encouragement, and technical support throughout the development of this project. Their valuable suggestions and feedback helped me understand the practical aspects of operating system hardening and security implementation.

Finally, I express my heartfelt gratitude to my parents, family, and friends for their constant encouragement, motivation, and support throughout the completion of this project.

This project has significantly enhanced my practical knowledge of Windows security mechanisms, PowerShell administration, endpoint protection, system monitoring, and cybersecurity best practices, which will greatly benefit my future academic and professional career.

ABSTRACT

Operating systems are a primary target for cyber threats such as malware, ransomware, unauthorized access, and network attacks. Operating system hardening is the process of improving system security by applying secure configurations and enabling built-in protection mechanisms to reduce vulnerabilities.

This project focuses on the practical hardening and security assessment of a Windows 11 operating system. Various security measures were implemented, including installing Windows security updates, verifying Microsoft Defender Antivirus, reviewing Windows Defender Firewall settings, managing user accounts and privileges, analyzing NTFS file permissions, enabling Controlled Folder Access for ransomware protection, assessing network configuration using PowerShell, reviewing Windows Event Logs, and creating a system restore point.

The implementation was carried out using native Windows security tools such as Windows Update, Microsoft Defender, Windows Defender Firewall, Windows PowerShell, Event Viewer, and System Protection. The security posture of the operating system was evaluated before and after applying the hardening measures, and the results were documented with practical observations and screenshots.

Table of Contents

1. Introduction	6
2. Objectives	8
3. Literature Review	9
4. Methodology	10
5. Practical Implementation	12
5.1. System Information and Initial Assessment	12
5.2. Windows Update and Patch Management	12
5.3. Microsoft Defender Antivirus Configuration	13
5.4. Windows Defender Firewall Verification	15
5.5. User Account Security	15
5.6. NTFS File System Security	16
5.7. Network Security Assessment	18
5.8. Event Monitoring and Logging	19
5.9. Backup and Recovery	20
5.10. Final Security Assessment	21
6. Results and Discussion	22
7. Conclusion	24
8. Future Scope	24
9. References	25

INTRODUCTION

1.1 Introduction

An operating system (OS) is the core software that manages computer hardware, software resources, and provides essential services for applications. It acts as an interface between users and computer hardware, enabling efficient execution of programs, memory management, file handling, process scheduling, and communication between system components. Popular operating systems include Microsoft Windows, Linux, and macOS, each providing different features and security mechanisms.

Operating system hardening is the process of improving the security of an operating system by reducing its attack surface and implementing security best practices. This includes installing security updates, configuring firewall settings, enabling antivirus protection, restricting user privileges, protecting sensitive files, monitoring system activities, and maintaining reliable backup and recovery mechanisms. A properly hardened operating system is more resistant to cyber threats and unauthorized access.

The project provides hands-on experience with practical security controls commonly used in modern Windows environments. It also demonstrates how applying multiple layers of security can significantly improve the overall security posture of an operating system.

1.2 Problem Statement

Modern operating systems face numerous security threats due to increasing internet connectivity, software vulnerabilities, and sophisticated cyberattacks. Many users operate their systems with default security settings without understanding the associated risks. Failure to apply security updates, improper user account management, weak access control, disabled security features, and poor monitoring can expose systems to malware, ransomware, and unauthorized access.

1.3 Need for the Project

The increasing dependence on digital systems has made cybersecurity an essential requirement for both individuals and organizations. Operating system hardening helps reduce security vulnerabilities, protects sensitive information, and improves the reliability of computer systems.

The need for this project includes:

- To understand practical operating system security.
- To learn Windows security features.
- To reduce system vulnerabilities.
- To improve protection against malware and ransomware.
- To gain hands-on experience with PowerShell and Windows administration.
- To implement industry-recommended security practices.

1.4 Scope of the Project

The scope of this project is limited to securing a standalone Windows 11 system using native Windows security features. The project includes:

- Windows Update and Patch Management
- Microsoft Defender Antivirus
- Windows Defender Firewall
- User Account and Privilege Management
- NTFS File System Security
- Controlled Folder Access
- Network Security Assessment
- Event Monitoring and Logging
- System Restore and Recovery

1.5 Significance of the Project

This project demonstrates how practical operating system hardening can improve the security of a Windows system using built-in security mechanisms. It also provides practical knowledge of Windows administration, PowerShell, endpoint protection, and security monitoring. The techniques implemented in this project can be applied in educational environments, personal computers, and small organizations to improve overall system security.

OBJECTIVES

2.1 General Objective

The primary objective of this project is to enhance the security of a Windows 11 operating system by implementing practical operating system hardening techniques using native Windows security features and PowerShell. The project aims to reduce security vulnerabilities, strengthen system protection, and improve the overall security posture by applying recommended cybersecurity best practices.

2.2 Specific Objectives

The specific objectives of this project are:

- To study the concept and importance of operating system hardening.
- To update the Windows operating system with the latest security patches.
- To verify and configure Microsoft Defender Antivirus for malware protection.
- To examine and verify Windows Defender Firewall settings.
- To implement the Principle of Least Privilege by reviewing local user accounts and administrative privileges.
- To analyze NTFS file system permissions and strengthen file security.
- To enable and verify Controlled Folder Access for ransomware protection.
- To assess network security using PowerShell commands.

2.3 Expected Outcomes

After completing this project, the following outcomes are expected:

- A Windows 11 system configured with enhanced security settings.
- Improved protection against common cyber threats such as malware, ransomware, and unauthorized access.
- Better understanding of Windows security features and administrative tools.
- Practical experience in using Windows PowerShell for security assessment and verification.
- Enhanced knowledge of operating system hardening techniques and cybersecurity best practices.

Literature Review

This chapter explains the concepts behind your implementation. It should be 2–3 pages and avoid unnecessary theory. I recommend dividing it into these sections:

3.1 Operating System Security

- Definition of operating system security
- Importance in modern computing

3.2 Operating System Hardening

- Definition
- Purpose
- Benefits
- Attack surface reduction

3.3 Common Cyber Threats

- Malware
- Ransomware
- Phishing
- Privilege Escalation
- Unauthorized Access

3.4 Security Principles

- CIA Triad
- Principle of Least Privilege (PoLP)
- Defense in Depth

3.5 Windows Security Features

- Windows Update
- Microsoft Defender Antivirus
- Windows Defender Firewall
- NTFS Permissions
- Event Viewer
- System Restore
- Controlled Folder Access

METHODOLOGY

4.1 Project Methodology

This project follows a practical approach to improve the security of a Windows 11 operating system using built-in Windows security features and administrative tools. The methodology involves identifying the system configuration, implementing security controls, verifying each security setting, and documenting the results with observations and screenshots.

The implementation was carried out in a sequential manner to ensure that each security feature was properly configured and validated before moving to the next stage.

4.2 Methodology Flow

The project was completed using the following steps:

1. System Analysis

- Gathered system information using Windows Settings and PowerShell.
- Verified Windows version and system specifications.

2. Patch Management

- Installed the latest Windows security updates and driver updates.
- Verified installed updates.

3. Endpoint Protection

- Updated Microsoft Defender Antivirus.
- Performed a Quick Scan.
- Verified real-time protection and security status.

4. Firewall Verification

- Checked Windows Defender Firewall settings.
- Verified Domain, Private, and Public profiles.

5. User Account Security

- Reviewed local user accounts.
- Verified administrator privileges.
- Confirmed unused accounts were disabled.

6. File System Security

- Examined NTFS permissions.

- Verified drive file system.
- Enabled Controlled Folder Access.

7. Network Security Assessment

- Reviewed IP configuration.
- Examined active network adapters.
- Checked active TCP connections.

8. Monitoring and Logging

- Reviewed Security and System logs using Event Viewer.
- Verified logging functionality.

9. Backup and Recovery

- Enabled System Protection.
- Created a System Restore Point.

10. Documentation

- Recorded PowerShell commands.
- Captured screenshots.
- Documented observations and security benefits.

4.3 Summary

The methodology ensures a structured implementation of Windows 11 security hardening by applying multiple layers of protection. Each configuration was verified using Windows utilities and PowerShell commands, and the results were documented to demonstrate the effectiveness of the implemented security measures.

PRACTICAL IMPLEMENTATION

5.1 System Information and Initial Assessment

Objective

The first step of operating system hardening is to identify the current system configuration. This helps verify the operating system version, hardware details, and existing security status before applying any security measures.

Procedure

System information was collected using Windows Settings and Windows PowerShell. The operating system version, edition, build number, processor details, installed memory, and computer name were verified.

PowerShell Commands Used

systeminfo

Get-ComputerInfo

Observation

The system was running Windows 11 Home Version 25H2 with the latest available build at the time of implementation. The collected information served as the baseline for the remaining security assessment.

5.2 Windows Update and Patch Management

Objective

To reduce security vulnerabilities by installing the latest Windows security updates and driver updates.

Procedure

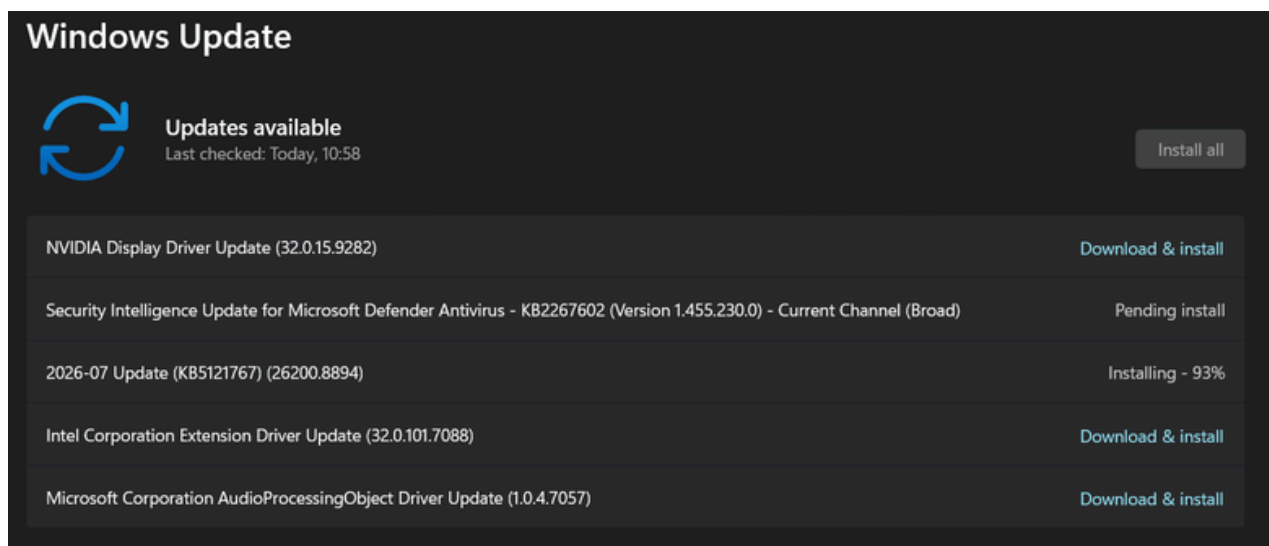
Windows Update was opened through the Settings application. Available cumulative updates, Microsoft Defender Security Intelligence updates, and device driver updates were installed. The installed updates were verified using PowerShell.

PowerShell Command

Get-HotFix | Sort-Object InstalledOn -Descending

Observation

The latest Windows cumulative updates and security patches were successfully installed. Updating the operating system ensured protection against known vulnerabilities.



5.1 Windows Updates

```
PS C:\Users\AADITYA> Get-HotFix | Sort-Object InstalledOn -Descending
```

Source	Description	HotFixID	InstalledBy	InstalledOn
AADI	Update	KB5121767	NT AUTHORITY\SYSTEM	20-07-2026 00:00:00
AADI	Security Update	KB5120102	NT AUTHORITY\SYSTEM	15-07-2026 00:00:00
AADI	Update	KB5100998	NT AUTHORITY\SYSTEM	15-07-2026 00:00:00
AADI	Update	KB5095182	NT AUTHORITY\SYSTEM	01-07-2026 00:00:00
AADI	Update	KB5054156	NT AUTHORITY\SYSTEM	30-10-2025 00:00:00

5.2 Installed Updates

5.3 Microsoft Defender Antivirus Configuration

Objective

To verify that Microsoft Defender Antivirus was fully enabled and capable of protecting the system against malware.

Procedure

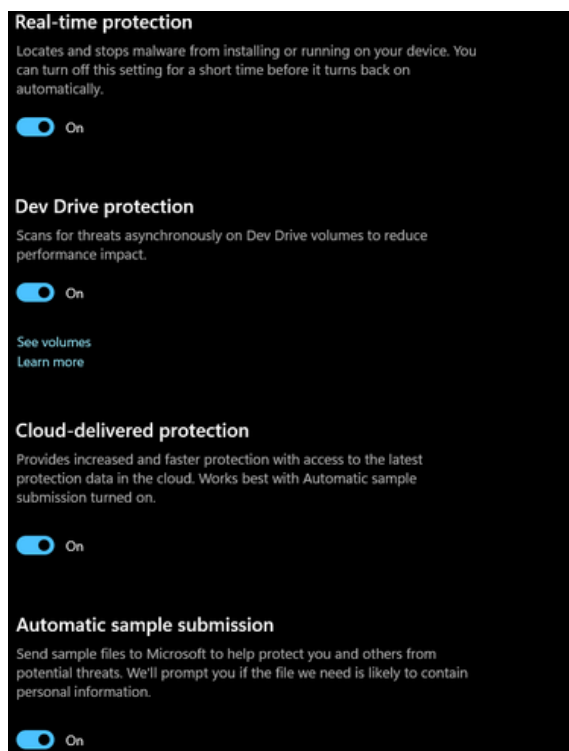
Microsoft Defender signatures were updated, followed by a Quick Scan. The antivirus configuration was verified using PowerShell.

PowerShell Commands

Update-MpSignatureStart-MpScan -ScanType QuickScanGet-MpComputerStatus

Observation

The antivirus engine was active with real-time protection, behavior monitoring, network inspection, and tamper protection enabled. Virus definitions were updated successfully, and no immediate threats were detected.



5.3 Windows Security Dashboard

```
PS C:\Users\AADITYA> Get-MpComputerStatus | Select-Object
>> AMServiceEnabled,
>> AntivirusEnabled,
>> AntispywareEnabled,
>> RealTimeProtectionEnabled,
>> BehaviorMonitorEnabled,
>> IoavProtectionEnabled,
>> NISEnabled,
>> IsTamperProtected,
>> AntivirusSignatureVersion,
>> AntivirusSignatureLastUpdated

AMServiceEnabled           : True
AntivirusEnabled           : True
AntispywareEnabled         : True
RealTimeProtectionEnabled  : True
BehaviorMonitorEnabled     : True
IoavProtectionEnabled      : True
NISEnabled                 : True
IsTamperProtected          : True
AntivirusSignatureVersion  : 1.455.234.0
AntivirusSignatureLastUpdated : 20-07-2026 08:49:15
```

5.4 PowerShell Output-

```
Start-MpScan -ScanType QuickScan
0/1 completed
[
Microsoft Defender Antivirus is scanning your device
This might take some time, depending on the type of scan selected.

Quick Scan
```

5.5 Quick Scan

5.4 Windows Defender Firewall Verification

Objective

To ensure the firewall was enabled for all network profiles.

Procedure

Firewall profiles were examined using Windows Security and PowerShell.

PowerShell Command

Get-NetFirewallProfile

Observation

The Domain, Private, and Public firewall profiles were enabled, ensuring protection against unauthorized network access.

```
PS C:\Users\AADITYA> Get-NetFirewallRule | Where-Object Enabled -eq True | Select-Object DisplayName, Direction, Action | Select-Object -First 20
```

DisplayName	Direction	Action
Wi-Fi Direct Spooler Use (Out)	Outbound	Allow
Network Discovery (SSDP-Out)	Outbound	Allow
Network Discovery (WSD Events-Out)	Outbound	Allow
Wi-Fi Direct Spooler Use (In)	Inbound	Allow
Remote Assistance (TCP-Out)	Outbound	Allow
Core Networking - Packet Too Big (ICMPv6-Out)	Outbound	Allow
Connected Devices Platform (UDP-Out)	Outbound	Allow
Core Networking - IPv6 (IPv6-In)	Inbound	Allow
Connected Devices Platform - Wi-Fi Direct Transport (TCP-Out)	Outbound	Allow
mDNS (UDP-Out)	Outbound	Allow
Delivery Optimization (UDP-In)	Inbound	Allow
Core Networking - Parameter Problem (ICMPv6-Out)	Outbound	Allow
Core Networking - Router Advertisement (ICMPv6-In)	Inbound	Allow
Core Networking - Destination Unreachable Fragmentation Needed (ICMPv4-In)	Inbound	Allow
Core Networking - Dynamic Host Configuration Protocol (DHCP-In)	Inbound	Allow
Network Discovery (WSD-In)	Inbound	Allow
Core Networking - Dynamic Host Configuration Protocol for IPv6(DHCPv6-In)	Inbound	Allow
Network Discovery for Teredo (SSDP-In)	Inbound	Allow
Connected Devices Platform (TCP-Out)	Outbound	Allow
Wi-Fi Direct Scan Service Use (In)	Inbound	Allow

5.6 Firewall Rules

5.5 User Account Security

Objective

To verify local user accounts and administrative privileges in accordance with the Principle of Least Privilege.

PowerShell Commands

Get-LocalUser

Get-LocalGroupMember Administrators

whoami

Observation

The built-in Administrator and Guest accounts were disabled. Only the primary user account possessed administrative privileges.

```
PS C:\Users\AADITYA> Get-LocalUser
```

Name	Enabled	Description
----	-----	-----
AADITYA	True	
Administrator	False	Built-in account for administering the computer/domain
DefaultAccount	False	A user account managed by the system.
Guest	False	Built-in account for guest access to the computer/domain
WDAGUtilityAccount	False	A user account managed and used by the system for Windows Defender Application Guard scen...

5.7 Local Users

```
PS C:\Users\AADITYA> Get-LocalGroupMember Administrators
```

ObjectClass	Name	PrincipalSource
-----	----	-----
User	AADI\AADITYA	MicrosoftAccount
User	AADI\Administrator	Local

5.8 Administration Group

5.6 NTFS File System Security

Objective

To verify the file system type and analyze NTFS permissions for secure file access.

PowerShell Command

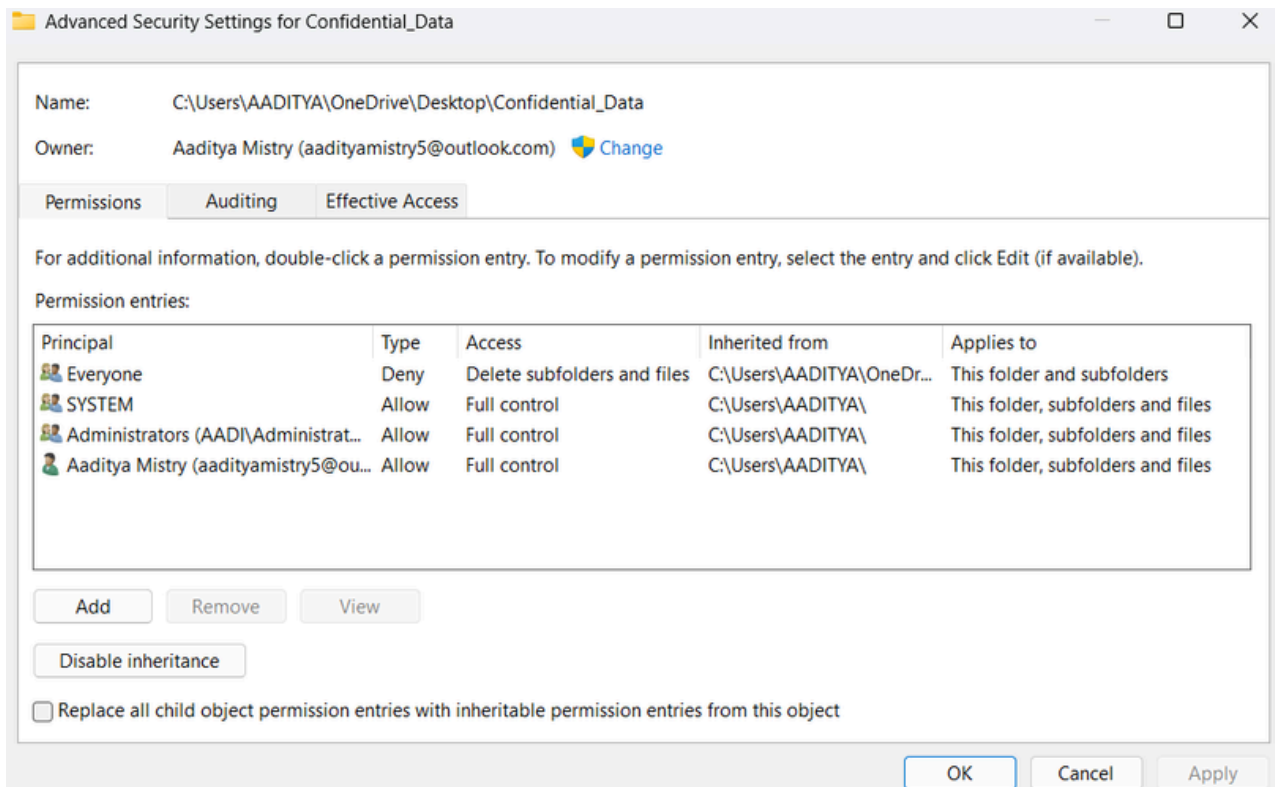
Get-Volume

Procedure

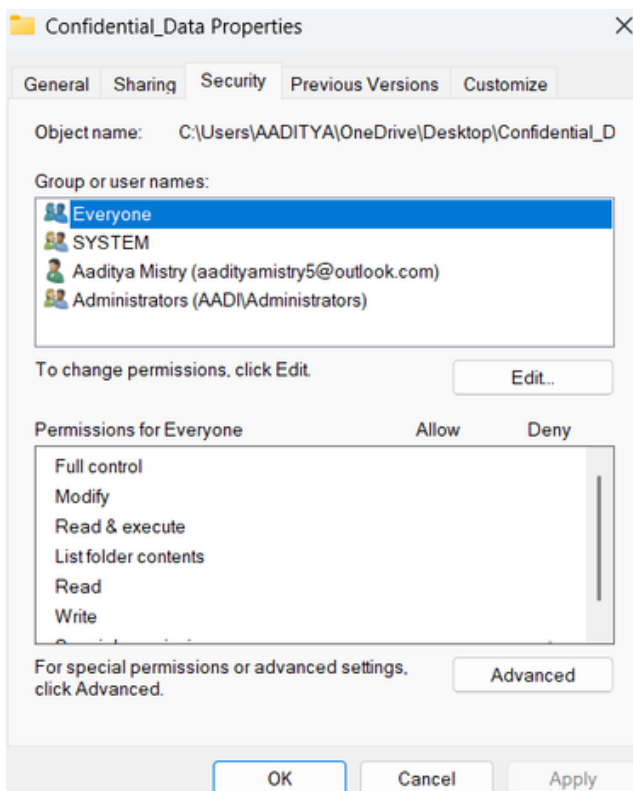
A confidential folder was created, and its NTFS permissions were reviewed using the Security tab in Windows Explorer.

Observation

The drives used the NTFS file system. Access permissions were assigned appropriately, and Controlled Folder Access was enabled to provide additional protection against ransomware.



5.9 NTFS Volume Information



5.10 Folder Permission



5.11 Controlled Folder Access

5.7 Network Security Assessment

Objective

To analyze network configuration and identify active network connections.

PowerShell Commands

```
ipconfig /all  
Get-NetAdapterNetstat -ano  
Get-NetTCPConnection  
Get-DnsClientServerAddress
```

Observation

The network configuration, DNS settings, network adapters, and active TCP connections were successfully examined to verify secure network communication.

```
C:\Users\AADITYA>ipconfig /all  
  
Windows IP Configuration  
  
Host Name . . . . . : Aadi  
Primary Dns Suffix . . . . . :  
Node Type . . . . . : Hybrid  
IP Routing Enabled. . . . . : No  
WINS Proxy Enabled. . . . . : No  
  
Ethernet adapter Ethernet 2:
```

5.12 IP Configurations

```
PS C:\WINDOWS\system32> Get-NetTCPConnection | Select-Object -First 20
```

LocalAddress	LocalPort	RemoteAddress	RemotePort	State	AppliedSetting	OwningProcess
::	65524	::	0	Bound		12296
::	65490	::	0	Bound		12296
::	63959	::	0	Bound		12296
::	63830	::	0	Bound		12296
::	63507	::	0	Bound		16084
::	63254	::	0	Bound		12296
::	63135	::	0	Bound		16084
::	63051	::	0	Bound		12296
::	62947	::	0	Bound		12296
::	60375	::	0	Bound		12296
::	59800	::	0	Bound		1524
::	58891	::	0	Bound		12296
::	57709	::	0	Bound		12296
::	57386	::	0	Bound		12296
::	57347	::	0	Bound		12296
::	56913	::	0	Bound		23228
::	56883	::	0	Bound		1524
::	56809	::	0	Bound		12296
::	56663	::	0	Bound		12296
::	55758	::	0	Bound		1524

5.13 Active TCP Connections-

5.8 Event Monitoring and Logging

Objective

To verify that Windows security events were being logged properly.

PowerShell Commands

```
Get-EventLog -LogName Security -Newest 20
```

```
Get-EventLog -LogName System -Newest 20
```

Observation

Security and System logs contained records of authentication events, system activities, and service operations, demonstrating that event logging was functioning correctly.

```
PS C:\WINDOWS\system32> Get-EventLog -LogName Security -Newest 20
```

Index	Time	EntryType	Source	InstanceID	Message
1305822	Jul 22 10:13	SuccessA...	Microsoft-Windows...	4798	A user's local group membership was enumerated....
1305821	Jul 22 10:13	SuccessA...	Microsoft-Windows...	4798	A user's local group membership was enumerated....
1305820	Jul 22 10:13	SuccessA...	Microsoft-Windows...	4798	A user's local group membership was enumerated....
1305819	Jul 22 10:13	SuccessA...	Microsoft-Windows...	4798	A user's local group membership was enumerated....
1305818	Jul 22 10:13	SuccessA...	Microsoft-Windows...	4798	A user's local group membership was enumerated....
1305817	Jul 22 10:13	SuccessA...	Microsoft-Windows...	4798	A user's local group membership was enumerated....
1305816	Jul 22 10:12	SuccessA...	Microsoft-Windows...	5379	Credential Manager credentials were read....
1305815	Jul 22 10:12	SuccessA...	Microsoft-Windows...	5379	Credential Manager credentials were read....
1305814	Jul 22 10:12	SuccessA...	Microsoft-Windows...	5379	Credential Manager credentials were read....
1305813	Jul 22 10:12	SuccessA...	Microsoft-Windows...	5379	Credential Manager credentials were read....
1305812	Jul 22 10:12	SuccessA...	Microsoft-Windows...	5379	Credential Manager credentials were read....
1305811	Jul 22 10:12	SuccessA...	Microsoft-Windows...	5379	Credential Manager credentials were read....
1305810	Jul 22 10:12	SuccessA...	Microsoft-Windows...	5379	Credential Manager credentials were read....
1305809	Jul 22 10:12	SuccessA...	Microsoft-Windows...	5379	Credential Manager credentials were read....
1305808	Jul 22 10:12	SuccessA...	Microsoft-Windows...	5379	Credential Manager credentials were read....
1305807	Jul 22 10:12	SuccessA...	Microsoft-Windows...	5379	Credential Manager credentials were read....
1305806	Jul 22 10:12	SuccessA...	Microsoft-Windows...	5379	Credential Manager credentials were read....
1305805	Jul 22 10:12	SuccessA...	Microsoft-Windows...	5379	Credential Manager credentials were read....
1305804	Jul 22 10:12	SuccessA...	Microsoft-Windows...	5379	Credential Manager credentials were read....
1305803	Jul 22 10:12	SuccessA...	Microsoft-Windows...	5379	Credential Manager credentials were read....

5.14 Security Logs

```
PS C:\WINDOWS\system32> Get-EventLog -LogName System -Newest 20
```

Index	Time	EntryType	Source	InstanceID	Message
40788	Jul 22 10:14	Information	Microsoft-Windows...	16	The description for Event ID '16' in Source 'Mi...
40787	Jul 22 10:14	Information	Service Control M...	1073748864	The start type of the Background Intelligent Tr...
40786	Jul 22 10:10	Information	Service Control M...	1073748864	The start type of the Background Intelligent Tr...
40785	Jul 22 10:09	Information	Microsoft-Windows...	16	The description for Event ID '16' in Source 'Mi...
40784	Jul 22 10:09	Information	Microsoft-Windows...	16	The description for Event ID '16' in Source 'Mi...
40783	Jul 22 10:09	Information	Microsoft-Windows...	16	The description for Event ID '16' in Source 'Mi...
40782	Jul 22 10:00	Information	Microsoft-Windows...	19	Installation Successful: Windows successfully i...
40781	Jul 22 10:00	Information	Microsoft-Windows...	43	Installation Started: Windows has started insta...
40780	Jul 22 10:00	Information	Microsoft-Windows...	44	Windows Update started downloading an update.
40779	Jul 22 09:53	Information	Service Control M...	1073748864	The start type of the Background Intelligent Tr...
40778	Jul 22 09:46	Information	Microsoft-Windows...	19	Installation Successful: Windows successfully i...
40777	Jul 22 09:46	Information	Microsoft-Windows...	43	Installation Started: Windows has started insta...
40776	Jul 22 09:46	Information	Microsoft-Windows...	19	Installation Successful: Windows successfully i...
40775	Jul 22 09:46	Information	Microsoft-Windows...	43	Installation Started: Windows has started insta...
40774	Jul 22 09:46	Information	Microsoft-Windows...	44	Windows Update started downloading an update.

5.15 System Logs

5.9 Backup and Recovery

Objective

To create a restore point for system recovery in case of security incidents or configuration failures.

Procedure

System Protection was enabled, and a restore point was created.

PowerShell Command

Get-ComputerRestorePoint

Observation

A restore point was successfully created, allowing the system to be restored to a previous stable state if required.

```
PS C:\WINDOWS\system32> Get-ComputerRestorePoint
```

CreationTime	Description	SequenceNumber	EventType	RestorePointType
22-07-2026 10:22:54	OS Hardening Project	1	BEGIN_SYSTEM_C...	16

5.16 Restore Point

5.10 Final Security Assessment

Summary of Implemented Security Measures

Security Feature	Status
Windows Updates	Updated
Microsoft Defender	Enabled
Firewall	Enabled
User Account Security	Verified
NTFS Permission	Verified
Controlled Folder Access	Enabled
Network Assessment	Completed
Event Logging	Verified
System Restore	Created

Observation

The implementation of multiple security controls significantly improved the security posture of the Windows 11 operating system. The system became more resilient against common cyber threats through regular updates, endpoint protection, secure user management, firewall enforcement, file system security, and monitoring mechanisms.

RESULTS AND DISCUSSION

6.1 Results

The practical implementation of operating system hardening was successfully completed on a Windows 11 system using built-in Windows security tools and PowerShell. All planned security measures were implemented and verified. The system was evaluated before and after the hardening process to ensure that the security configurations were functioning correctly.

Security Measure	Status	Outcome
Windows Security Updates	Completed	Latest updates installed
Microsoft Defender Antivirus	Enabled	Real-time protection active
Windows Defender Firewall	Verified	All network profiles enabled
User Account Security	Verified	Built-in accounts disabled, administrator privileges reviewed
NTFS File System Security	Completed	Secure file permissions verified
Controlled Folder Access	Enabled	Ransomware protection activated
Network Security Assessment	Completed	Network configuration and active connections analyzed
Event Monitoring	Completed	Security and system logs verified
System Restore	Created	Restore point successfully generated

6.2 Discussion

The implementation demonstrated that Windows 11 provides several effective built-in security features that can significantly improve system protection when properly configured. Installing the latest security updates reduced exposure to known vulnerabilities, while Microsoft Defender Antivirus provided continuous protection against malware and other malicious software.

The verification of NTFS permissions and the activation of Controlled Folder Access strengthened file security and provided additional protection against ransomware attacks. Network analysis and event log monitoring confirmed that the system was functioning correctly and that security-related activities were being recorded for future analysis.

Finally, the creation of a system restore point improved system recoverability by allowing restoration to a previous stable configuration if required. Overall, the practical implementation showed that combining multiple security controls provides a layered defense, making the operating system more resilient against common cyber threats.

6.3 Challenges Faced

During the implementation of this project, a few challenges were encountered:

- Windows 11 Home edition does not support BitLocker encryption, limiting disk encryption options.
- Some PowerShell commands required administrative privileges.
- Understanding firewall rules and event logs required careful analysis.
- Security settings were verified using built-in tools rather than enterprise management solutions.

These challenges were addressed using alternative Windows security features and by carefully validating each configuration step.

6.4 Key Learning Outcomes

Through this project, the following practical skills were gained:

- Understanding of operating system hardening concepts.
- Experience with Windows PowerShell for system administration.
- Knowledge of Microsoft Defender and Windows Firewall configuration.
- Understanding of NTFS permissions and file security.
- Ability to perform basic network security assessment.
- Experience in security monitoring using Event Viewer.
- Knowledge of backup and recovery through System Restore.
- Improved documentation and technical reporting skills.

CONCLUSION

7.1 Conclusion

This project successfully demonstrated the practical implementation of operating system hardening techniques on a Windows 11 system using built-in Windows security features and PowerShell. The primary objective of enhancing system security through native security mechanisms was achieved by implementing multiple layers of protection.

Throughout the project, several security measures were applied, including installing the latest Windows updates, verifying Microsoft Defender Antivirus, reviewing Windows Defender Firewall settings, managing user accounts and privileges, analyzing NTFS file permissions, enabling Controlled Folder Access, assessing network security, monitoring system logs, and creating a system restore point. Each implementation was verified using Windows administrative tools and PowerShell commands, with observations documented to evaluate their effectiveness.

The project demonstrated that a well-configured operating system is significantly more resistant to common cyber threats such as malware, ransomware, unauthorized access, and network-based attacks. It also provided valuable hands-on experience in Windows system administration, cybersecurity best practices, and technical documentation. Overall, this project highlights the importance of regularly maintaining and hardening an operating system to improve its security, reliability, and resilience against evolving cyber threats.

FUTURE SCOPE

Although the project successfully implemented essential operating system hardening techniques, several advanced security measures can be incorporated in future work to further strengthen system protection. Potential enhancements include:

- Implementing BitLocker Drive Encryption on Windows Pro or Enterprise editions.
- Configuring Group Policy for centralized security management.
- Integrating Active Directory for enterprise user and access management.

- Deploying Security Information and Event Management (SIEM) tools for advanced log monitoring and threat detection.
- Performing regular vulnerability assessments using professional security tools.
- Implementing multi-factor authentication (MFA) to strengthen user authentication.
- Applying advanced endpoint protection and Endpoint Detection and Response (EDR) solutions.
- Automating security monitoring and reporting using PowerShell scripts.

These enhancements would provide a more comprehensive security framework suitable for enterprise environments and further improve the overall security posture of the operating system.

REFERENCES

Use a consistent citation style (APA or IEEE). Here are suitable references for your project:

1. Microsoft. Windows Security Documentation.
<https://learn.microsoft.com/windows/security>
2. Microsoft. Microsoft Defender Antivirus Documentation.
<https://learn.microsoft.com/defender-endpoint>
3. Microsoft. Windows Defender Firewall Documentation.
<https://learn.microsoft.com/windows/security/operating-system-security/network-security/windows-firewall>
4. Microsoft. Windows PowerShell Documentation.
<https://learn.microsoft.com/powershell>
5. Microsoft. Windows Update Documentation.
<https://learn.microsoft.com/windows/deployment/update>
6. William Stallings, Operating System Security, Pearson Education.
7. Andrew S. Tanenbaum & Herbert Bos, Modern Operating Systems, 4th Edition, Pearson.
8. Charles P. Pfleeger, Security in Computing, Pearson.